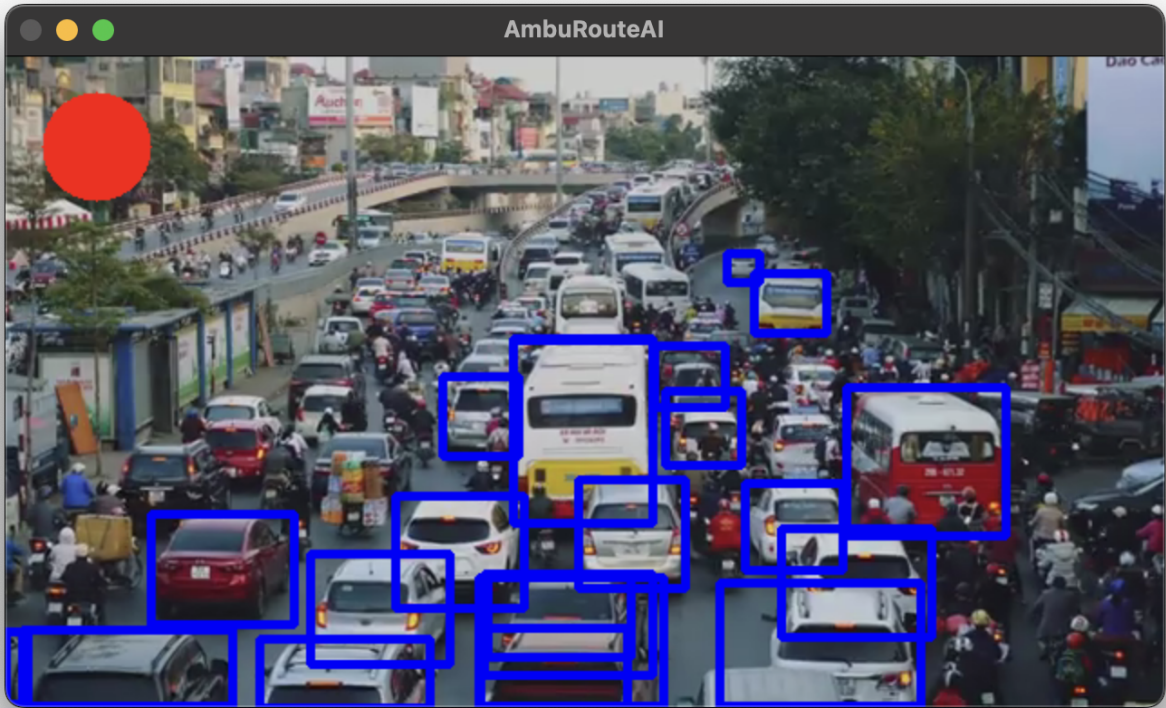


AmbuRouteAI



 An AI-Powered Traffic Management for Emergency Vehicles.

 AmbuRouteAI is an AI-powered traffic management system that detects ambulances in real-time and dynamically controls traffic signals to provide a clear path for emergency vehicles. Using YOLOv8 for object detection and OpenCV for traffic light simulation, this system helps reduce ambulance response time and enhances emergency healthcare logistics.

Features

- ✔ **Real-time Ambulance Detection** – Uses YOLOv8 AI to identify ambulances from live traffic feeds. ✔
- Smart Traffic Signal Control** – Dynamically changes traffic lights using OpenCV when an ambulance is detected. ✔
- ✔ **Live Video Processing** – Works with webcams or traffic camera feeds to analyze road conditions. ✔
- ✔ **Seamless Integration** – Can be extended to IoT-enabled smart city infrastructure. ✔
- Scalable & Efficient** – Future-ready for integration with Google Maps API for real-time route optimization.

Tech Stack

Component	Technology/Tool
AI Model	YOLOv8 (Ultralytics)

Component	Technology/Tool
Computer Vision	OpenCV, NumPy
Backend	Python
Traffic Simulation	OpenCV
Live Video Input	Webcam / CCTV Feed

How It Works

- 1 AI-Powered Detection** – YOLOv8 identifies ambulances in real-time from traffic video feeds.
- 2 Traffic Light Control** – If an ambulance is detected, the traffic light turns green; otherwise, it remains red.
- 3 Visual Alerts & UI** – The system draws bounding boxes around detected ambulances and simulates traffic signals.
- 4 Extensibility** – The project can integrate with Google Maps API & IoT sensors for smarter city-wide traffic control.

Installation & Setup

Run

Download [yolov8n.pt](#)

```
git clone https://github.com/mantreshkhurana/AmbuRouteAI.git
cd AmbuRouteAI
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python main.py -p traffic_video.mp4 # Use an MP4 file
python main.py -l # For live webcam feed
```

Future Enhancements

- ☐ **Integrate with Google Maps API** – Real-time traffic data for route optimization.
- ☐ **IoT-Based Smart Traffic Signals** – Connect with Raspberry Pi & Arduino for real-world applications.
- ☐ **Hospital Alert System** – Send emergency notifications to hospitals about incoming patients.
- ☐ **Mobile App Interface** – Develop an ambulance tracking app with live traffic updates.



Authors

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